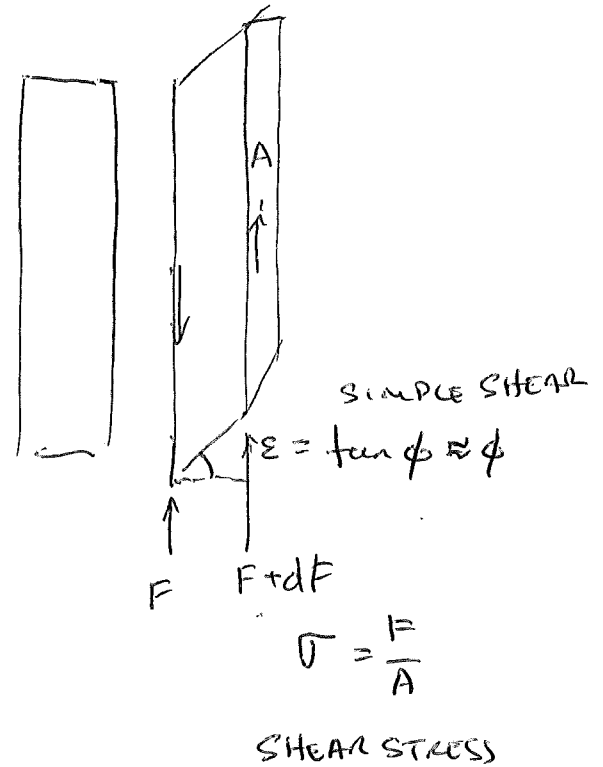
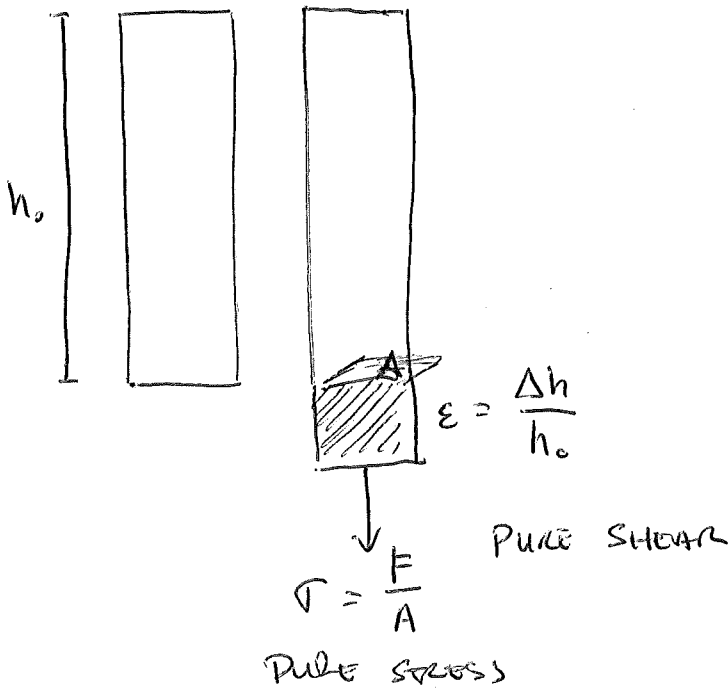
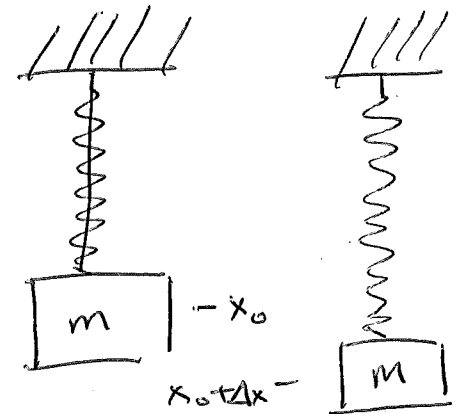
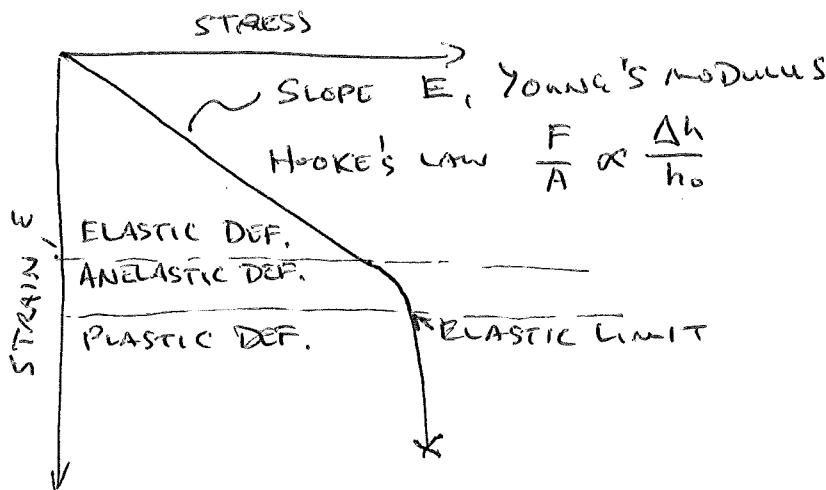


STRESS STRAIN



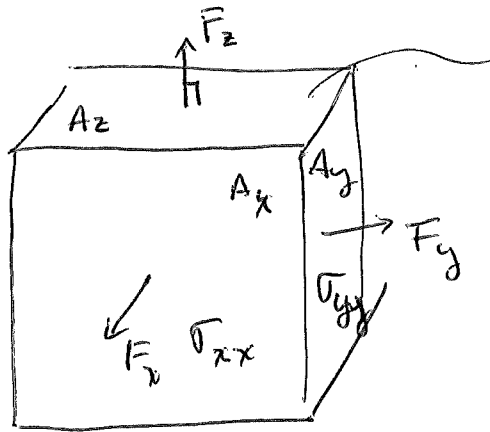
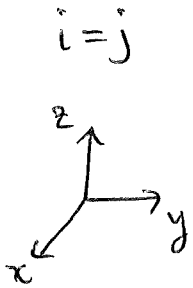
$$\sigma = \frac{F}{A} \propto \frac{\Delta h}{h_0} = \epsilon$$



$F = k \Delta x$ HOOKE'S LAW
 RESTORING FORCE $\uparrow$ SPRING CONSTANT

(2)

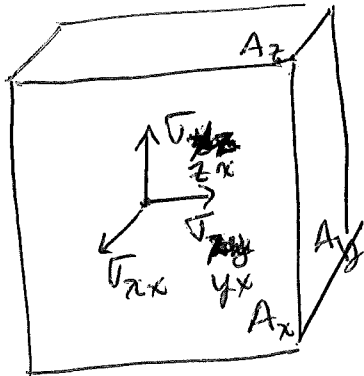
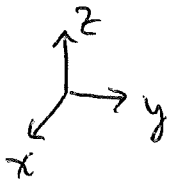
$$\sigma_{ij} = \lim_{A_j \rightarrow 0} \frac{F_i}{A_j}$$



$$\sigma_{zz} = \lim_{A_z \rightarrow 0} \frac{F_z}{A_z}$$

NORMAL STRESS
"PURE"

$i \neq j$



$$\sigma_{ij} = \lim_{A_j \rightarrow 0} \frac{F_i}{A_j}$$

SHAPE
SHEAR STRESS

u - DISPLACEMENT FIELD

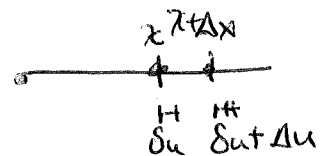
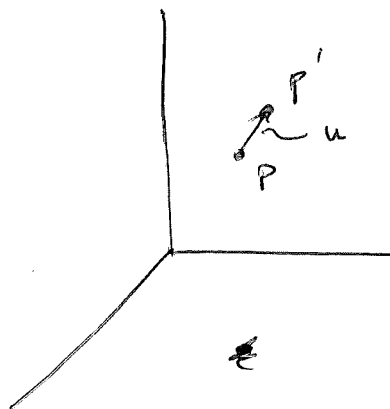
$$u = u(x, y, z)$$

NORMAL STRAIN

$$\epsilon_{xx} = \frac{\partial u_x}{\partial x}$$

$$\epsilon_{yy} = \frac{\partial u_y}{\partial y}$$

$$\epsilon_{zz} = \frac{\partial u_z}{\partial z}$$



$$\frac{\Delta V}{V} = \Delta = \epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz}$$

DILATION

SCALAR DISPLACEMENT

POTENTIAL

$$\Delta = \nabla \cdot u$$

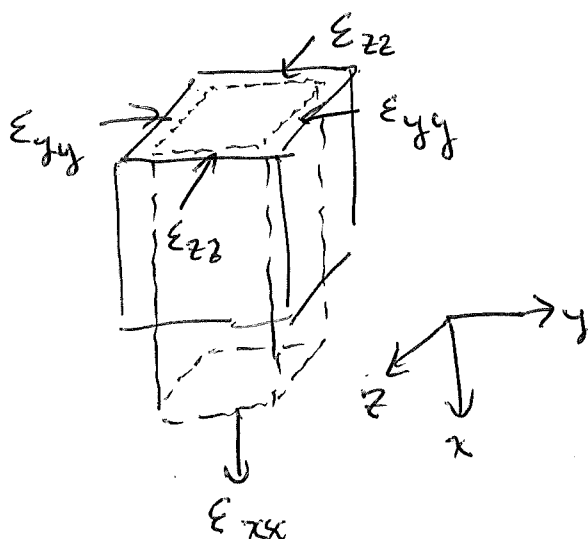
$$\frac{\partial^2 \Delta}{\partial t^2} = \alpha^2 \nabla^2 \Delta$$

↑

WAVE VELOCITY

P-WAVE

$$V_p = \alpha = \sqrt{\frac{K + \frac{4}{3}\mu}{\rho}}$$



K - BULK MODULUS [Pa]

μ - SHEAR MODULUS [Pa]

ρ - DENSITY [kg m^{-3}]

$$\epsilon_{xx} = -\nu \epsilon_{yy}$$

$$\epsilon_{xx} = -\nu \epsilon_{zz}$$

}

ISOTROPIC

SAME PROPERTIES IN ALL DIMENSIONS

ν - POISSON'S RATIO

$$\sigma_{xx} = E \epsilon_{xx}$$

E = YOUNG'S MODULUS

~~$$E = \frac{3K\mu}{1+\nu}$$~~

$$E = 3K(1-2\nu)$$

$$E = 2\mu(1+\nu)$$

For a Poisson solid, $\nu = \frac{1}{4}$. Most rocks, $0.25 < \nu < 0.3$

$\nu \approx 0.25$.

$$\frac{V_p}{V_s} = \left(\frac{2 \left(1 - \frac{1}{4} \right)}{1 - 2 \left(\frac{1}{4} \right)} \right)^{1/2}$$
$$= \sqrt{3}$$

So to time an earthquake

$$\tau_i = \frac{t_{s_i} - t_{p_i}}{(\sqrt{3} - 1)} V_p$$

V_p of crystalline rocks.

GRANITE

DIOBASE

GABBRO

PERIDOTITE 7.9-8.2 km/s

ECLOGITE 7.9-8.2 km/s

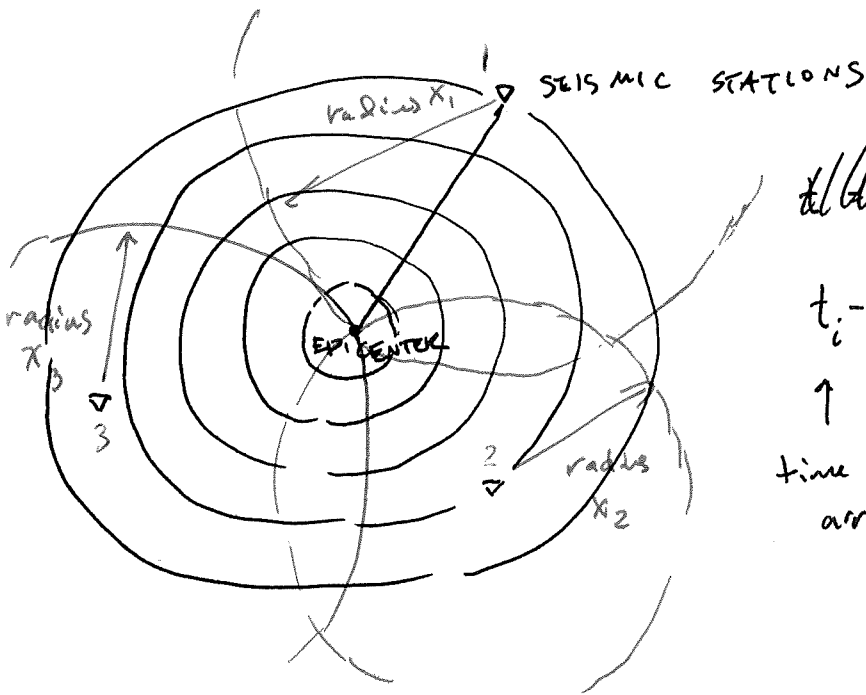
> are discriminate by
Poisson's number

SHEAR WAVES

$$\frac{\partial^2 \vec{\psi}}{\partial t^2} = \beta^2 \nabla^2 \vec{\psi}$$

$$V_s = \beta = \sqrt{\frac{\mu}{\rho}} \quad \text{SHEAR VELOCITY}$$

LOCATING EARTHQUAKES



$$t_i - t_0 = \frac{x_i}{v}$$

distance from epicenter of station i

time of earthquake

time of arrival at station i

t_0 is unknown.

How to find t_0 ,

V_p , V_s TRAVEL AT DIFFERENT VELOCITIES

↑ ↑

Fast slow

$$V_p > V_s$$

$$t_{p,i} - t_0 = \frac{x_i}{V_p} \quad (1)$$

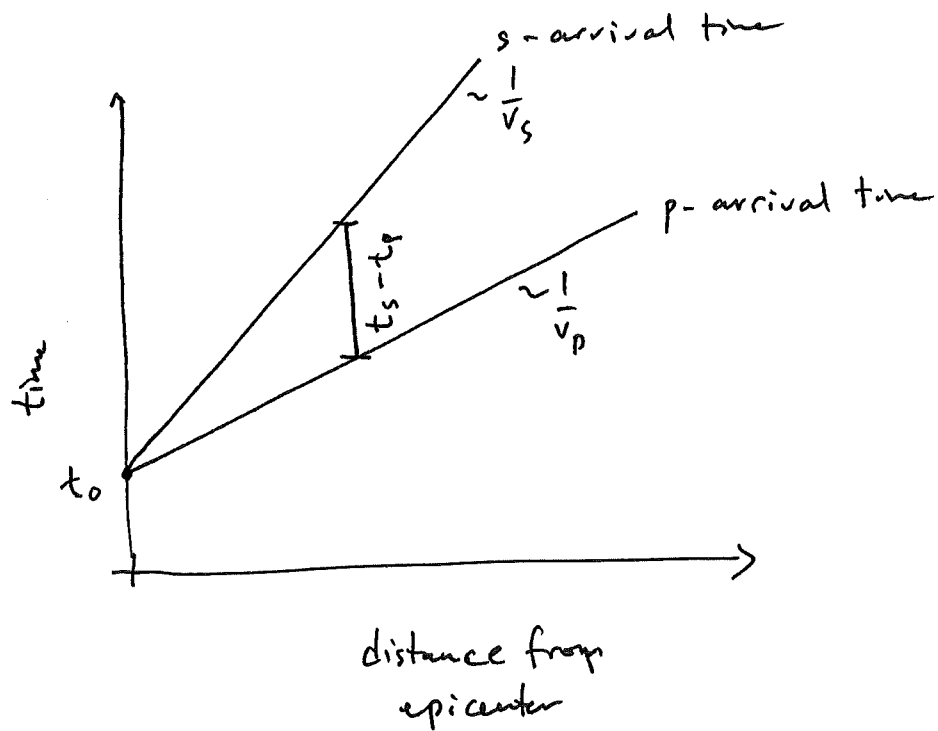
$$t_{s,i} - t_0 = \frac{x_i}{V_s} \quad (2)$$

$$(2) - (1)$$

$$t_{s,i} - t_{p,i} = \frac{x_i}{V_s} - \frac{x_i}{V_p}$$

$$t_{s,i} - t_{p,i} = x_i \left(\frac{1}{V_s} - \frac{1}{V_p} \right) = x_i \left(\frac{V_p}{V_s V_p} - 1 \right) \frac{1}{V_p}$$

$$x_i = \frac{t_{s,i} - t_{p,i}}{\left(\frac{V_p}{V_s} - 1 \right)} V_p$$



WHAT IS $\frac{v_p}{v_s}$

$$v_p = \sqrt{\frac{K + \frac{4}{3}\mu}{\rho}}$$

$$K = \frac{E}{3(1-2\nu)}$$

$$v_s = \sqrt{\frac{\mu}{\rho}}$$

$$\mu = \frac{E}{2(1+\nu)}$$

can show

$$\frac{v_p}{v_s} = \left(\frac{2(1-\nu)}{1-2\nu} \right)^{1/2}$$

$$\begin{aligned} \left(\frac{v_p}{v_s} \right)^2 &= \left(\frac{K + \frac{4}{3}\mu}{\cancel{\rho}} \right) \left(\frac{\cancel{\rho}}{\mu} \right) \\ &= \frac{K}{\mu} + \frac{4}{3} \end{aligned}$$

$$\left(\frac{v_p}{v_s} \right)^2 = \frac{\cancel{E}}{3(1-2\nu)} \cdot \frac{2(1+\nu)}{\cancel{E}} + \frac{4}{3}$$

$$= \frac{2(1+\nu) + 4(1-2\nu)}{3(1-2\nu)}$$

$$= \frac{6 - 6\nu}{3(1-2\nu)}$$

$$= \frac{6(1-\nu)}{3(1-2\nu)}$$

$$= \frac{2(1-\nu)}{1-2\nu}$$

$$\frac{v_p}{v_s} = \left(\frac{2(1-\nu)}{1-2\nu} \right)^{1/2}$$